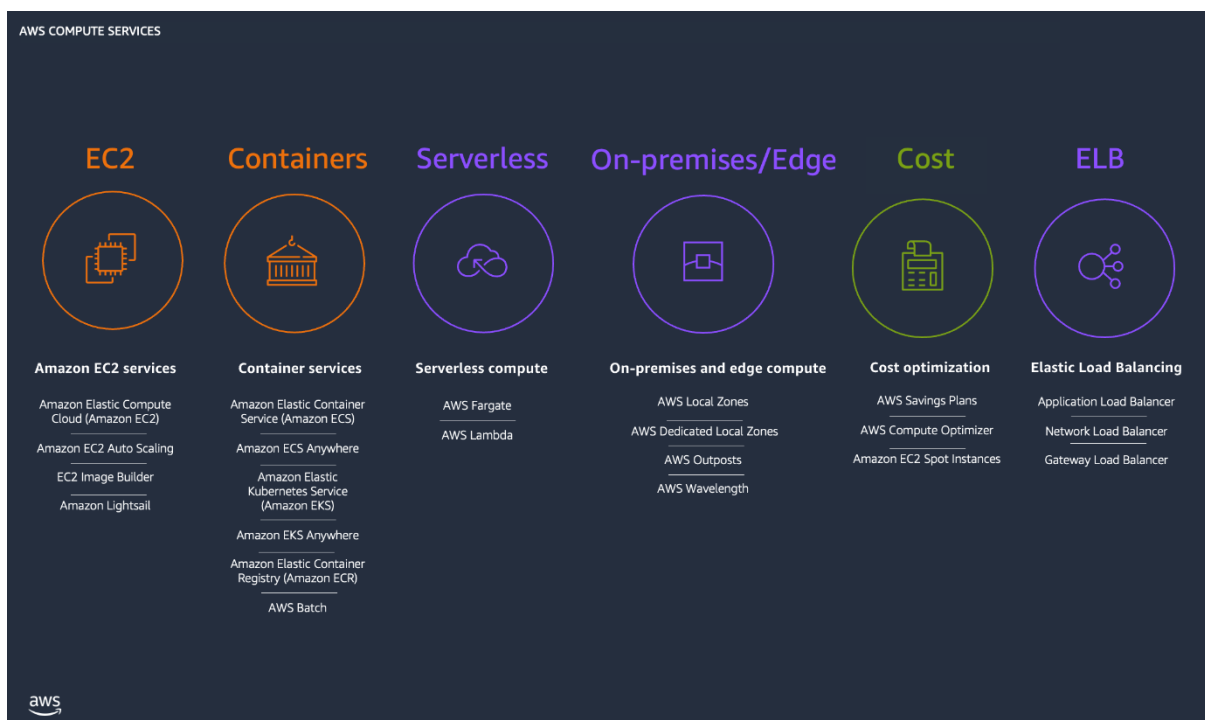


AWS Networking Computer service-42

Introduction

- AWS provides a broad set of **networking** and **compute** services that enable secure, scalable, and high-performance applications.
- Networking services connect resources, manage traffic, and ensure secure communication.
- Compute services provide the infrastructure to run workloads ranging from web apps to machine learning.

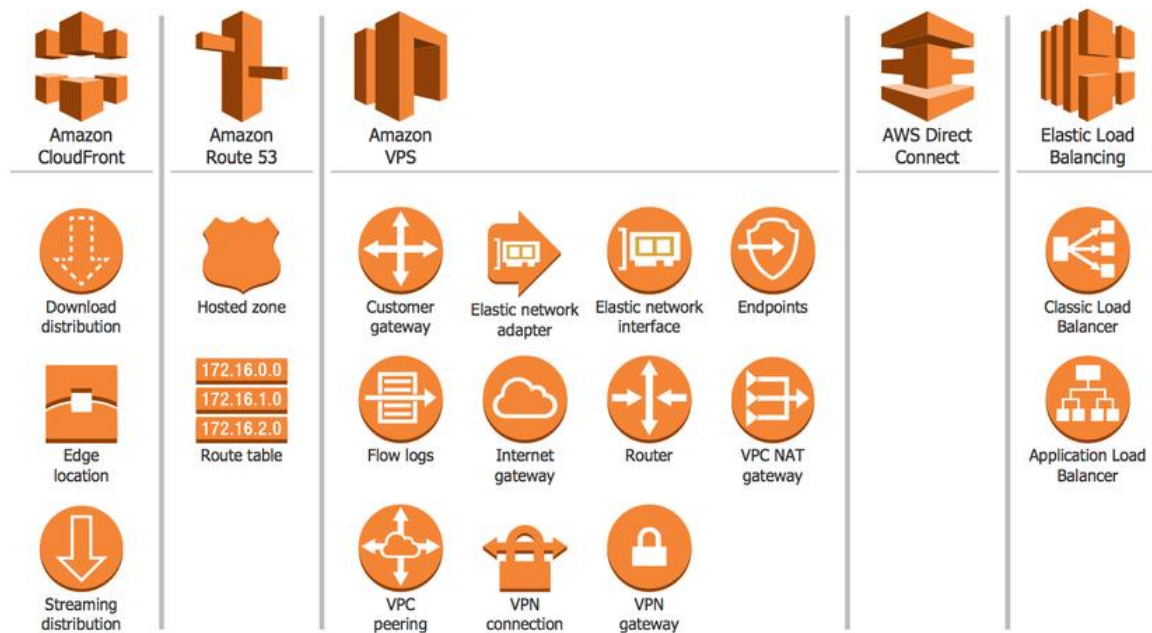


AWS Networking Services

- **Amazon VPC (Virtual Private Cloud)**
 - Creates isolated virtual networks in AWS.
 - Supports subnets (public and private), route tables, and internet/NAT gateways.
 - Enables fine-grained network segmentation and security control.
- **Elastic Load Balancing (ELB)**
 - Distributes incoming traffic across multiple targets like EC2, containers, or IPs.

- Types: Application Load Balancer (ALB), Network Load Balancer (NLB), Gateway Load Balancer (GWLB).
 - Improves fault tolerance and availability.
- **Amazon Route 53**
 - Highly available DNS service.
 - Provides domain registration, DNS routing, and health checks.
 - Supports latency-based, failover, and geolocation routing policies.
- **AWS Direct Connect**
 - Dedicated network connection between on-premises data centers and AWS.
 - Provides consistent low-latency performance.
 - Often used for hybrid cloud environments.
- **AWS Transit Gateway**
 - Central hub for connecting multiple VPCs and on-premises networks.
 - Simplifies large-scale networking and reduces management overhead.
- **AWS CloudFront (Content Delivery Network)**
 - Distributes content globally with low latency.
 - Caches data at edge locations.
 - Integrates with S3, EC2, and other AWS services.
- **AWS Global Accelerator**
 - Improves application availability and performance using AWS global network.
 - Directs traffic to the optimal endpoint based on latency, geography, and health.
- **AWS VPN**
 - Securely connects on-premises networks to AWS VPCs.
 - Options include Site-to-Site VPN and Client VPN.

Networking & Content Delivery



AWS Compute Services

- **Amazon EC2 (Elastic Compute Cloud)**
 - Scalable virtual servers in the cloud.
 - Provides a wide range of instance types optimized for compute, memory, storage, or GPU workloads.
 - Supports autoscaling for elasticity.
- **Amazon ECS (Elastic Container Service)**
 - Fully managed container orchestration service.
 - Runs Docker containers on EC2 or AWS Fargate.
 - Integrates with IAM and VPC for security.
- **Amazon EKS (Elastic Kubernetes Service)**
 - Managed Kubernetes service.
 - Provides Kubernetes cluster management without operational overhead.
 - Enables hybrid and multi-cloud portability.
- **AWS Lambda**
 - Serverless compute service.

- Runs code without provisioning or managing servers.
- Charges only for execution time and requests.
- Commonly used for event-driven applications.
- **AWS Fargate**
 - Serverless compute engine for containers.
 - Works with ECS and EKS.
 - Eliminates the need to manage EC2 instances.
- **Amazon Lightsail**
 - Simplified cloud compute for small applications, blogs, and websites.
 - Provides preconfigured virtual servers with predictable pricing.
- **AWS Outposts**
 - Extends AWS infrastructure to on-premises data centers.
 - Provides consistent hybrid cloud experience.
- **AWS Batch**
 - Runs batch computing workloads at scale.
 - Dynamically provisions compute resources based on job requirements.
- **Amazon EMR (Elastic MapReduce)**
 - Big data processing framework.
 - Runs Apache Hadoop, Spark, and Presto on AWS infrastructure.

Example Use Case

- A company hosts a **web application** using:
 - **EC2** instances for compute.
 - **Elastic Load Balancer** for traffic distribution.
 - **VPC** with public and private subnets for isolation.
 - **RDS** database in private subnet.
 - **CloudFront** for global content delivery.

- **Direct Connect** for hybrid connectivity with their on-premises data center.

Benefits of AWS Networking and Compute Services

- High scalability with auto-scaling and elastic networking.
- Global reach with AWS's worldwide infrastructure.
- Security through IAM, VPC isolation, and encryption.
- Cost efficiency with pay-as-you-go and serverless options.
- Flexibility to choose from VMs, containers, or serverless compute.

Conclusion

- AWS networking services provide the backbone for connectivity, security, and performance.
- AWS compute services offer multiple models to run workloads, from EC2 to serverless Lambda.
- Together, they enable businesses to build highly available, secure, and scalable cloud applications.